Mnemos-DNA: Gated Delta-Rule Linear Attention with k-mer Fast-Weight Memory for Edge-Scale DNA Modeling

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Abstract

Gated delta-rule linear attention (Kimi Delta Attention, KDA) has strong natural-language results but, to our knowledge, no genomic evaluation. We study it at an edge-scale operating point — ∼2.3M-parameter character-level models, one 8 GB consumer GPU, pretraining on GRCh38 at a 131,072 bp context — where single-axis ablations are affordable to run to completion, and add a per-8-mer *fast-weight memory* (**Mnemos-DNA**): 65,536 content-addressed 16×16 delta-rule matrices beside each KDA layer. (i) **Recall.** On a repeat-copy probe over real hg38 background, a fully matched pure-KDA control shows no measurable copy advantage beyond 64 bp; neither do a windowed local-attention hybrid, pure KDA with 4× or 16× the recurrent state, nor a same-size table indexed by position instead of content. The k-mer memory yields 0.53–0.87 nats of copy advantage out to 64 kbp and retains most of it under the tested 1–3 point substitutions per repeat. (ii) **Cost, and a dial for it.** It adds 2.9% parameters but a 4096× larger recurrent inference state (128 KiB → 512 MiB) and ≈2× prefill cost: the price of recall is paid in state, not parameters. A sweep of the key length $k \in \{4, 6, 7, 8\}$ turns this into a dial: recall reach grows monotonically with the address space and fades once the repeat distance approaches $\approx 4^k$ bp, so a k=7 table keeps 71% of the k=8 advantage at 64 kbp with a quarter of the state. (iii) **A negative result.** Under one matched fine-tuning protocol, pretrained Mnemos-DNA is *worse* than its own random initialization on splice sites (0.740 vs. 0.832); a local window plus length curriculum (**Mnemos-DNA-C**) does not repair this, and the released 1M-context HyenaDNA checkpoint shows the same signature. We also show that fine-tuning-protocol drift alone produces spurious cross-arm effects of up to 30 points, an order of magnitude larger than any architecture effect we measure, while perplexity separates the arms by ≤0.01 nats. Our reading: in long-context DNA modeling, exact recall, context length and short-range grammar are separate budgets.

1 Introduction

DNA foundation models face a context problem that natural-language models do not: regulatory interactions span hundreds of kilobases, yet single-nucleotide resolution matters, so a sequence of interest is routinely 10^5–10^6 tokens long. The two families that dominate this regime — HyenaDNA's implicit long convolutions [1] and the Mamba-based Caduceus [2] — replace quadratic attention with sub-quadratic mixers. A third family, *gated delta-rule linear attention* (Gated DeltaNet [3], Kimi Delta Attention [4]), has strong natural-language results but no published genomic evaluation. Its defining weakness is also its most interesting property for genomics: the recurrent state is

Code and evaluation harness: https://github.com/B4K2/Mnemos-DNA. Pretrained checkpoints (all four arms): https://huggingface.co/B4K2xx/Mnemos-DNA.

a fixed $d \times d$ matrix, so how much exact recall of distant motifs it can sustain is capacity-limited, a recall–capacity trade-off formalized in prior work [5]. Genomes, full of near-exact repeats whose *differences* carry the signal, should be a stress test for exactly this bound.

Edge-scale is the design objective, not a constraint we apologize for. This work targets an operating point of ∼2.3M parameters and a single 8 GB consumer GPU (RTX 5060 Laptop). That choice is deliberate and has two independent justifications. First, it is a deployment regime with its own demand: sequencing-adjacent analysis increasingly runs on laptops, clinic workstations, and field devices, where a model that processes a megabase window in 5.6 GB is usable and a released checkpoint that goes out of memory at 512 kbp is not. Second, it is the regime in which a full single-axis ablation is actually finishable: every arm here is pretrained from scratch on a matched tokenizer, matched data order and matched token budget, which is what licenses attributing a difference to the one ingredient that changed. Large-scale reports rarely can, because they bundle architecture, budget, and data changes. We accordingly do not claim that findings here transfer to 100M+ scale, and we mark the places where the d_{model}=128 ceiling, rather than an architecture, is what a number measures.

Questions. (i) Is KDA a viable base operator for a character-level DNA model at 131 kbp training context? (ii) Where does its finite-state recall bound show up on genomic tasks, and can it be repaired within an $O(N)$ time budget — and at what cost in state? (iii) What does an ultra-long fixed pretraining context do to sub-kilobase downstream capability?

Contributions. The primary contribution is the first two items (recall bound, memory, and its cost); the splice-site degradation is a secondary finding, and protocol sensitivity a methodological one.

- **Mnemos-DNA, a k-mer fast-weight memory** (§3.1): a content-addressed memory in every layer, keyed by the exact last 8-mer, each address holding a 16×16 delta-rule fast-weight matrix [6]. It substantially improves recall on our repeat-copy probe, with controls attributing the gain to content addressing, while staying $O(N)$ in time and bitwise-exact under chunked low-memory inference.
- **An explicit collision, capacity and memory-footprint analysis** (§3.2–§3.3): what the delta rule does under repeated writes to one slot, a pigeonhole bound on collisions in a 131 kbp window, the measured departure from that bound on real hg38, and a byte-level table showing that a ∼3% parameter increase buys a 4096× increase in recurrent inference state. A key-length sweep (k=4–8, §4.7) measures the resulting state–recall trade-off directly and identifies its mechanism: overwrite by intervening occurrences of the same address.
- **An applications evaluation, not only an architecture one** (§4): pretraining, single-nucleotide-resolution classification (Genomic Benchmarks, Nucleotide Transformer), species identification across input lengths, long-range regulatory and variant-effect prediction, and the 31-task GUE suite — each with its own reading, including where our arms lose.
- **A protocol-sensitivity result for fine-tuned genomic benchmarks** (§4.2, §4.5). We quantify, on the same checkpoints, how much cross-arm variation in fine-tuning protocol (pooling head, epoch budget, training-set caps) distorts architecture comparisons: up to 30 points per task and 5 points per suite mean — spurious effects an order of magnitude larger than the real cross-arm differences, which are ≤1 point per suite under a single frozen protocol

with 2–3 seeds. Against that frozen protocol, a matched random-init control beats *both* pretrained arms by ∼9 points on splice sites: fixed ultra-long-context pretraining measurably damages sub-kilobase positional grammar, and the window-plus-curriculum patch designed to repair it does not.

- **A same-harness baseline comparison, framed as deployment feasibility** (§4.10): released HyenaDNA and Caduceus checkpoints run through our identical harness at inference batch 1. They are ∼3× our parameter count (6.5–7.7M) with orders-of-magnitude larger pretraining budgets, and are run out-of-the-box without a chunked path, so this measures what an 8 GB deployment of released artifacts gets — not an intrinsic algorithmic advantage over Hyena or Mamba.

2 Preliminaries and Related Work

DNA sequence models. Character-level genomic language models trade the compression of k-mer or BPE tokenization for single-nucleotide resolution, which SNP- and splice-site-level tasks require. HyenaDNA [1] pretrains implicit-convolution models up to 1 M context on hg38; Nucleotide Transformer [7] and DNABERT-2 [8] take the tokenized-transformer route at much larger scale; Caduceus [2] is a bidirectional, reverse-complement-equivariant Mamba masked LM at 131 K context. We adopt HyenaDNA's tokenizer and held-out-chromosome split so that our evaluation rows are readable against theirs, and compare against *released* checkpoints through our harness, never re-implementations.

Long-context strategies. Implicit long convolutions [9], selective state-space models [10] and linear attention [11] all reduce the softmax transformer's [12] quadratic term, but they differ in what they keep: a convolution keeps a filter over relative offsets, an SSM keeps a decayed state, linear attention keeps an outer-product memory. What none of them keeps is an unbounded exact record of what was seen.

Kimi Delta Attention. KDA [4] is a linear-attention recurrence with a channel-wise forget gate, per-head write strength β, ℓ_2-normalized queries and keys, and a short causal convolution on $q/k/v$. Each head maintains a $d_v \times d_k$ state S_t (so that the retrieval $S_{t-1}k_t$ lands in value space, the correction $v_t - S_{t-1}k_t \in \mathbb{R}^{d_v}$ is well-typed, and the outer-product write $(v_t - S_{t-1}k_t)k_t^\top$ is $d_v \times d_k$) updated by a gated delta rule — an outer-product write that first subtracts what the key currently retrieves, then writes the new value — giving it error-correcting fast-weight semantics [6, 5] at $O(N)$ time and $O(1)$ state per head:

$$S_t \;=\; \mathrm{diag}(\alpha_t)\, S_{t-1} \;+\; \beta_t \left(v_t - S_{t-1}k_t\right)k_t^\top, \qquad \text{read } S_t q_t. \tag{1}$$

The recall bound. Because S_t has fixed size, the number of (key, value) associations recoverable exactly is bounded; past that, writes interfere. In language this surfaces on synthetic multi-query associative recall [5] and on string copying, where fixed-state models fall behind attention [13]; the repeated-token loss drop our probe measures is the signature of induction-style copying [14]. No established recall benchmark exists for genomic language models — HyenaDNA, Caduceus and Nucleotide Transformer report none — so we introduce one: a repeat-copy probe on real hg38 background (§4.6).

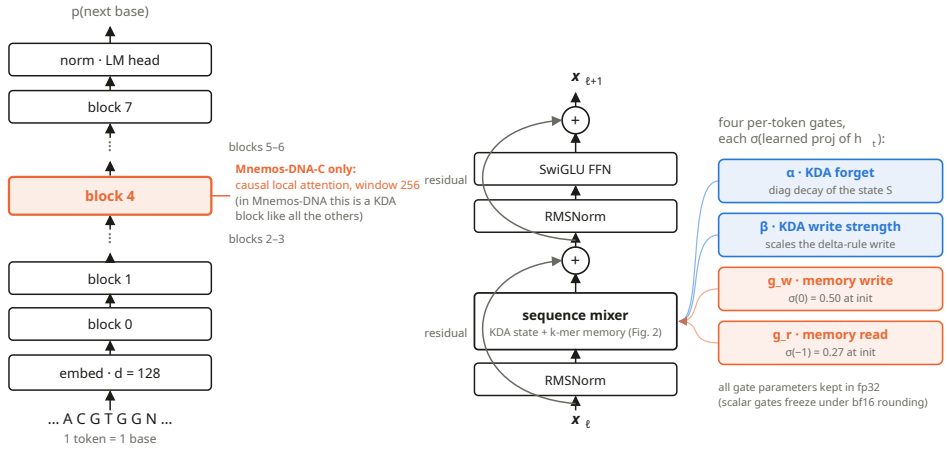


Figure 1: The Mnemos-DNA(-C) architecture, drawn vertically, input at the bottom. **(a)** Eight pre-norm blocks, d_{model}=128, 4 heads, no positional encoding (blocks 2–3 and 5–6 elided; they are identical to blocks 1 and 7). In Mnemos-DNA-C only, block 4's mixer becomes blocked causal local attention (window 256, effective span 256–511 bp). **(b)** One block's datapath: pre-norm residual structure (RMSNorm → mixer → residual, RMSNorm → SwiGLU → residual), and the four per-token gates that steer the mixer, each a sigmoid of a learned projection of the token's hidden state — the KDA recurrence's channel-wise forget gate α_t and write strength β_t, and the k-mer memory's write gate g_w (bias init 0, σ=0.50) and read gate g_r (bias init -1, $\sigma\approx$0.27). All gate parameters are kept in fp32; scalar gates freeze under bf16 rounding (§3.1). The mixer's internals are Figure 2.

3 Method

3.1 Mnemos-DNA (D06): KDA with per-k-mer fast-weight memory

The backbone is 8 layers, d_{model}=128, 4 heads, vocabulary of 8 (A/C/G/T/N plus specials), no positional encoding (NoPE), 2.32M parameters; Figure 1 shows the stack and one block's datapath with its four gates, and Figure 2 the mixer's two memories. Alongside each layer's KDA mixer sits a content-addressed memory keyed by the exact identity of the last k=8 nucleotides (4^8=65,536 addresses). Each address stores not a vector but a small delta-rule fast-weight matrix (d_k=d_v=16): when position t's 8-mer address is a, the layer writes (k_t, v_t) projections of the hidden state into slot a's matrix under a learned write gate g_w (bias init 0.0, σ=0.5),

$$W_a \;\leftarrow\; W_a + g_w\big(v - W_a k\big)k^\top, \tag{2}$$

and reads slot a's matrix with the current query under a learned read gate (bias init -1.0, $\sigma \approx 0.27$), adding $g_r \cdot W_a q$ to the mixer output before the output projection. Two properties matter:

- **It targets the recall bound directly.** Recall of "what followed this exact 8-mer last time" no longer competes for the KDA state's fixed capacity: the address space factors it out, and the delta rule inside each slot handles collisions of repeated 8-mers (§3.2).

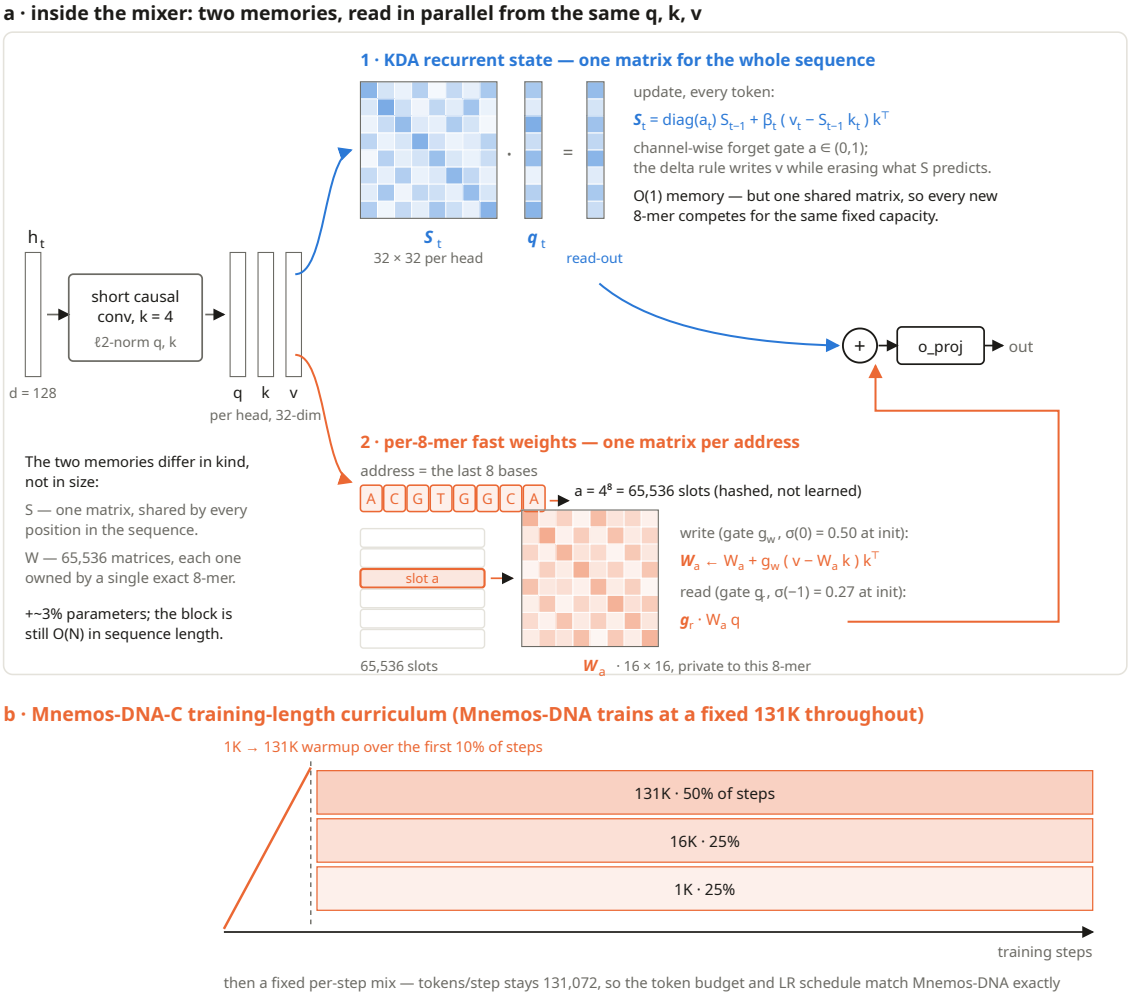


Figure 2: Inside the mixer: the two memories, and the training-length curriculum. **(a)** Two memories read in parallel from the same conv-smoothed q, k, v. *Top:* the KDA recurrence keeps one 32×32 state matrix S_t per head for the entire sequence and reads it as the matrix–vector product $S_t q_t$; the gated delta rule writes v_t while erasing what S_{t-1} already predicts, so every new k-mer competes for one fixed capacity. *Bottom:* the fast-weight memory instead holds 4^8=65,536 separate 16×16 matrices, one per exact 8-mer, and writes (g_w-gated) / reads (g_r-scaled) only the slot the current 8-mer addresses. The two reads are summed before the output projection. The table is reset per sequence and carried exactly across chunks at inference. **(b)** Mnemos-DNA-C's training-length curriculum; tokens per step are unchanged, so Mnemos-DNA and Mnemos-DNA-C share a token budget and learning-rate schedule.

- **It stays $O(N)$ in time and chunk-exact.** The full table is never materialized during the training forward — only chained per-chunk carries are computed — and chunked inference carries the table across chunks bitwise-exactly (verified at 16 K), which is what legitimizes the 1 Mbp low-memory evaluation path of §4.10. It is emphatically not free in *space*; §3.3 gives the bytes.

Two training details were load-bearing rather than incidental: gate parameters are kept in fp32 (scalar gates freeze under bf16 rounding), and the read/write gate initializations above came from a bias sweep after an earlier bug zeroed the write gate.

3.2 Collisions and capacity of the 8-mer address space

The address space is smaller than the sequences it indexes, so collisions are not an edge case — they are the normal operating regime, and the delta rule's behaviour under them is what determines whether the memory stores anything useful.

How many collisions. With $\Sigma = \{A, C, G, T\}$ there are exactly $4^8 = 65{,}536$ addresses. A window of $N = 131{,}072$ bp yields $N - k + 1$ addressed positions, so by the pigeonhole principle at least $N - 4^k = 65{,}536$ writes must land on an already-written slot, *whatever* the sequence distribution: a collision rate of at least 50%. Under a uniform i.i.d. null the expected number of distinct slots touched is $4^k\left(1 - e^{-N/4^k}\right) = 56{,}661$ (86.5% of the space), giving an expected collision rate of 56.8%. Genomic sequence is not uniform, and §4.8 measures the real figures: 41,342 slots touched (63.1%) and a 68.5% collision rate. The pigeonhole bound is therefore a floor that real DNA clears by a wide margin, and the uniform null *under*states collisions — the direction that matters.

What a collision does. Suppose slot a receives writes $(k_1, v_1), \ldots, (k_m, v_m)$ with gates $g_1, \ldots, g_m$ over one sequence, starting from $W_a^{(0)} = 0$ (the table is reset per sequence). Equation 2 gives

$$W_a^{(i)} \;=\; W_a^{(i-1)}\left(I - g_i k_i k_i^\top\right) \;+\; g_i\, v_i k_i^\top, \tag{3}$$

using $\|k_i\|_2 = 1$ (k is ℓ_2-normalized). Two regimes follow, and they are the two things the memory can be:

Proposition 1 (Exact overwrite along a repeated key)**.** *If successive writes to slot a share the same key, $k_i = k$ for all i, then along that key the slot is an exponential moving average of the written values: $W_a^{(i)}k = (1-g_i)\,W_a^{(i-1)}k + g_i v_i$. With a constant gate g, after m writes $W_a^{(m)}k = \sum_{i=1}^{m} g(1-g)^{m-i}v_i$, an EMA with effective horizon $1/g$: the slot remembers the last $\approx 1/g$ occurrences of that 8-mer and forgets earlier ones geometrically.*

Proposition 2 (Interference-free storage along orthogonal keys)**.** *If the written keys are mutually orthogonal, $k_i^\top k_j = 0$ for $i \neq j$, then $W_a^{(m)}k_j = g_j v_j$ for every $j \leq m$: all m associations coexist in the slot, up to the gate scaling. Since $W_a \in \mathbb{R}^{16\times 16}$, at most $d_k = 16$ mutually orthogonal keys fit; beyond that, writes project onto the span of existing keys and Proposition 1's overwrite behaviour applies along the overlapping directions.*

Equation 3 is the recursive-least-squares/delta-rule update restricted to one address, so the general case interpolates: the slot performs gated least-squares regression of v on k over the writes it receives, retaining up to 16 well-separated key directions exactly and blending the rest as a

Quantity	Size	Derivation
Parameters		
Backbone (KDA, FFN, embeddings)	2.25 M	—
k-mer memory projections	66.6 K	$8 \times (4{\cdot}128{\cdot}16 + 129)$
Mnemos-DNA total	2.32 M	+2.9% over the backbone
Recurrent state carried at inference (batch 1)		
KDA recurrent state	128 KiB	8 layers × 4 heads × 32×32 × 4 B
k-mer fast-weight table (Mnemos-DNA)	512 MiB	8 layers × 65,536 slots × 16×16 × 4 B
k-mer fast-weight table (Mnemos-DNA-C)	448 MiB	7 k-mer layers; block 4 is local attention
Ratio, table to KDA state	**4096×**	

Table 1: Parameter and inference-state accounting for the k-mer fast-weight memory. Fast-weight tables and the KDA state are kept in fp32 (4 B); the recurrence is run outside autocast because scalar gates freeze under bf16 rounding. **The headline asymmetry: parameters grow 2.9% while the recurrent state a deployment must carry grows 4096×, from 128 KiB to 512 MiB.** This is the true cost of factoring exact 8-mer recall out of the fixed KDA state, and it is paid mostly in space, plus a ≈2× prefill-throughput cost relative to the matched pure-KDA control (§4.10); throughput remains flat in sequence length. A bf16 table would halve the figure to 256 MiB at the precision risk noted above, and the key length is a second dial: the table is $8 \cdot 4^k$ KiB, and §4.7 measures what each factor of four buys in recall. During training the full table is never materialized (only chained per-chunk carries), which is what keeps a 131 kbp training step inside 8 GB; the 512 MiB is what a streaming inference deployment holds.

gate-weighted EMA. Two kinds of collision must therefore be kept apart. An *address collision* — two positions sharing an 8-mer — is what the 68.5% counts. *Representational interference* — their learned keys k_i being similar — is what actually overwrites. Address collisions are survivable only to the extent the keys separate: the 16-direction capacity above is an upper bound, not a guarantee, since keys are learned projections of hidden states and nothing forces them to be orthogonal. The analysis also predicts where the memory should fail: when one address receives many writes with *similar* keys and *different* values, only the most recent $\approx 1/g$ survive. This makes recall *distance*-dependent even though the window-level collision rate is not: what matters for recalling an occurrence at distance d is how many writes to the *same* address land in between, about $d/4^k$ under a uniform null. Recall should therefore fade once d approaches 4^k bp, shifted by genomic non-uniformity and by how well the intervening keys separate. §4.7 tests this prediction by varying k.

3.3 Memory footprint accounting

The parameter count understates what this memory costs at inference, so we state both explicitly. Table 1 gives the byte-level breakdown.

3.4 Mnemos-DNA-C (D07): a coupled prototype (local window + length curriculum)

Mnemos-DNA-C adds two things to Mnemos-DNA at once, each motivated by a specific observed failure (§4.2): a *randomly initialized* Mnemos-DNA fine-tuned on Nucleotide Transformer splice sites beats the 4.5B-token 131 K-context checkpoint (0.832 vs. 0.740 under the final matched protocol) — which we hypothesised was fixed-131 K pretraining tuning the recurrence's timescales *away* from 300–600 bp grammar (a hypothesis, not an established mechanism). As §4.2 reports, the patch did *not* in the end repair that failure; we describe the design here because the arm exists and its

Arm	Mixer	Params	Position	Train ctx	Extras
D01 (Pure-KDA)	KDA only	2.25M	none	131,072	matched control
D02	KDA + local attn (1-in-4 layers, w=512)	∼2.0M	NTK-RoPE	262,144	5.0B-token budget
D06 (Mnemos-DNA)	KDA + k-mer fast-weight memory	2.32M	none	131,072	—
D07 (Mnemos-DNA-C)	Mnemos-DNA, block 4 → local attn (w=256)	2.29M	none	131,072	length curriculum

Table 2: The arms analysed in this paper. D01 (Pure-KDA) is the KDA-only control, fully matched to Mnemos-DNA(-C): NoPE, 131 K training context, 4.5B tokens, same data order. (An earlier pure-KDA arm, D04, was excluded after an audit found its bf16 scalar cache gates had never trained; Appendix B.) D02 is the window-only control — windowed local attention *without* a k-mer memory — and appears in the language-model sweep (Figure 4), the recall probe (Figure 9), the long-range probes (Table 4) and the embedding analysis (Figure 7). D02 predates the Mnemos-DNA(-C) protocol freeze: it was trained with NTK-RoPE [15, 16] at a 262 K context for a 5.0B-token budget, so its rows are read as a window-only reference point, not as a member of the matched Mnemos-DNA(-C) pair. Other earlier ablation arms of the wider study (a kernelized-linear-attention arm, an evicted-KV cache, a vector-valued k-mer memory) established the design and are referenced where relevant.

other properties (recall, efficiency) are part of the matrix.

- **Local window.** Block 4's mixer becomes blocked causal local attention with window 256 (effective causal span 256–511 bp): exact token-level recall inside the window regardless of learned timescales. Window 256 rather than 512 was chosen so that a 300–600 bp task sequence is not resolvable in a single exact hop.
- **Length curriculum.** Training length warms up 1 K→131 K over the first 10% of steps, then draws per-step lengths from a fixed mix {1K: 0.25, 16K: 0.25, 131K: 0.5}. Tokens per step are unchanged (131,072), so the token budget and learning-rate schedule are identical to Mnemos-DNA. A loss-driven length controller was considered and rejected: bang-bang switching oscillates and is not reproducible across arms; the fixed mix keeps both regimes in the loss by construction, monitored by long- and short-context validation losses.

Mnemos-DNA-C is a coupled prototype, and we do not claim otherwise. Because the window and the curriculum were introduced together, any result attributed to Mnemos-DNA-C in this paper is a property of the pair. The single-axis controls exist as configurations — curriculum without window, and Mnemos-DNA itself (neither) — but the curriculum-only arm was untrained at this preprint's checkpoint. Under the matched fine-tuning protocol of §4.2 the pair turns out to change downstream means by ≤1 point, so the attribution question has become less urgent than initially assumed; what the pair *does* measurably change is long-range recall (slightly down, §4.6) and short-context validation loss (held within 0.03 nats of long-context, Figure 3).

Micro-batching is capped by tokens in flight, not sequences ($4 \times 32{,}768$ peaks at 6.17 GiB where $1 \times 131{,}072$ peaks at 4.35 GiB), which is what lets the curriculum train on the 8 GB card. Mnemos-DNA-C has 2.29M parameters.

4 Experiments

Pretraining setup. Human reference genome GRCh38/hg38 [17], single-nucleotide tokenization (no k-mers, no BPE; vocabulary 8), HyenaDNA's held-out-chromosome split, causal next-nucleotide objective. Token budget 4.5B (34,177 steps × 131,072 tokens), AdamW, peak learning rate 10^{-3} with cosine decay to 0.1×, warmup 200 steps, bf16 with gradient checkpointing. Hardware: one

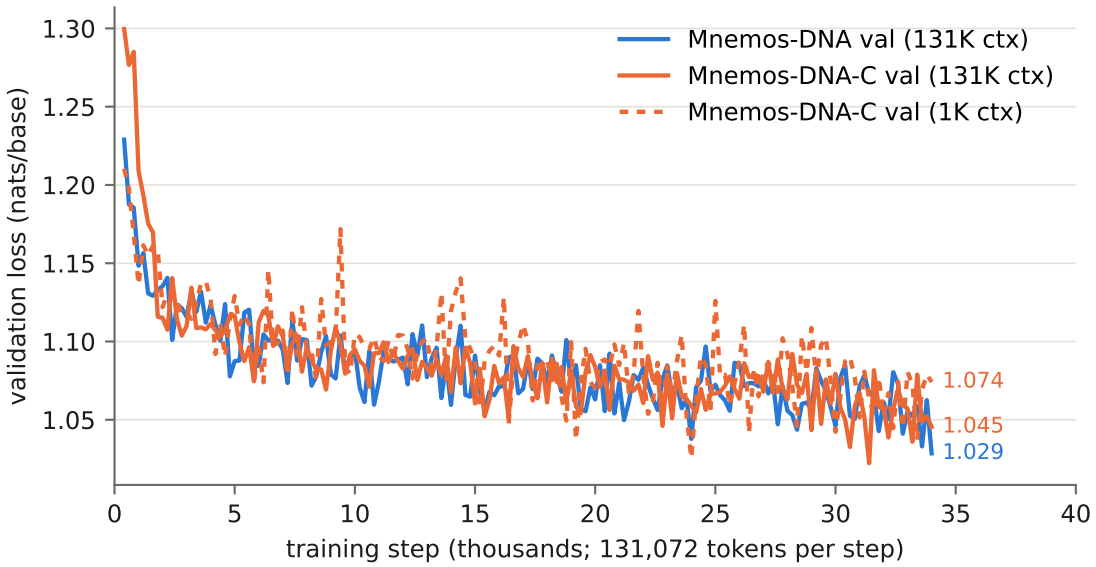


Figure 3: Validation loss. Mnemos-DNA and Mnemos-DNA-C track each other closely at the shared 131 K evaluation length (final 1.029 vs. 1.045 nats/base); Mnemos-DNA-C additionally holds a 1 K-context validation loss within 0.03 nats of its long-context loss, the curriculum's explicit target. Curves start after learning-rate warmup.

NVIDIA RTX 5060 Laptop GPU (8 GB, Blackwell). **The checkpoints analysed are trained on 4.5B tokens** — step 34,177 (Mnemos-DNA), 33,900 (Mnemos-DNA-C) and 34,408 (Pure-KDA, D01) — and every number below comes from those checkpoints.

Evaluation harness. Every number comes from one suite: per-(benchmark, context) reseeding of numpy and torch, a 4M-token floor for language-model evaluation per context length, 128 trials per needle point, and a percentile bootstrap (n=1000) on every test-set metric — resampling per prediction (classification), per window (LM; per-token bootstraps understate the interval by an order of magnitude), or per trial (needle). These intervals cover split-sampling uncertainty only, not seed variance. Downstream tasks run in two modes: a frozen-backbone linear probe (a lower bound, labelled as such) and full fine-tuning (the protocol family of HyenaDNA / Nucleotide Transformer / DNABERT-2; effective batch 32, up to 6 epochs / 6,000 steps, best-dev-epoch selection, a [mean | last | 8 positional bins] head pooled over forward and reverse-complement passes). External baselines — `hyenadna-large-1m`, `hyenadna-medium-450k`, `caduceus-ph/ps` (131 K) — run through the same harness with vocabulary remapped losslessly in both directions. Caduceus is a bidirectional masked LM, so causal LM and needle metrics are skipped for it by design. The chunked low-memory path is used only for our KDA arms, where it is verified bitwise-exact; the external checkpoints are run out-of-the-box, so their out-of-memory ceilings describe those released artifacts as shipped (§4.10).

4.1 Pretraining on the human genome

Validation loss tracks closely between the arms throughout training (Figure 3). At 4.5B tokens, held-out cross-entropy at 131 K context is 1.065 (Mnemos-DNA), 1.067 (Mnemos-DNA-C) and 1.074 (Pure-KDA, D01) nats/base — 1.54 bits/bp, perplexity 2.90 over the 4-letter alphabet — and stays within ±0.03 nats from 8 K to 1 M bp of evaluation context for all arms (Figure 4). The

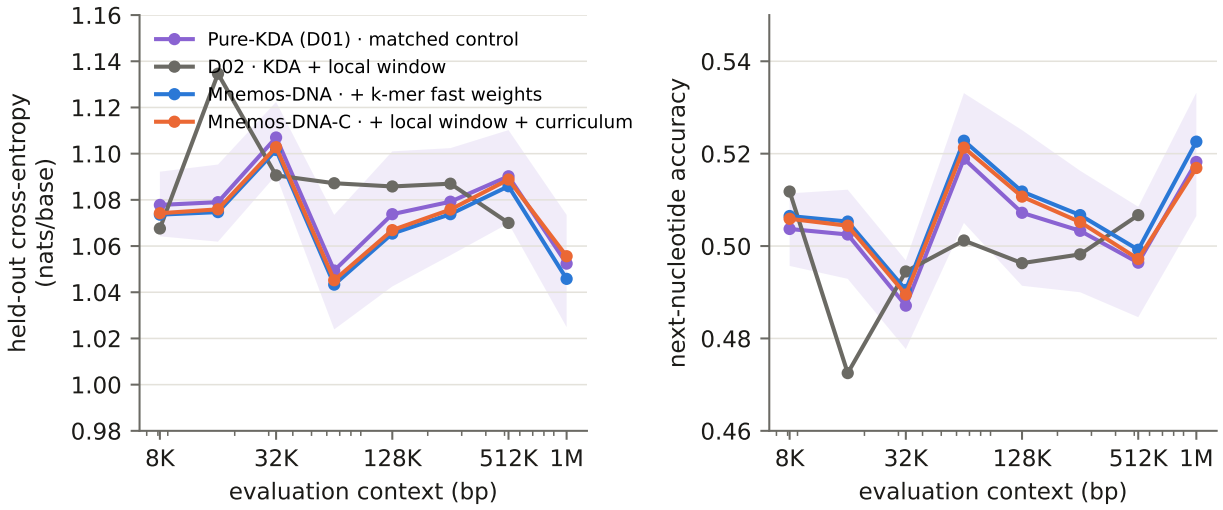


Figure 4: Held-out language-model quality does not degrade with evaluation context, from 8 kbp to 1 Mbp. Left: cross-entropy (a 4 M-token floor per point; scatter is genomic-window variation, not a trend). Right: next-nucleotide accuracy against a 0.25 chance rate. The k-mer-memory arms (Mnemos-DNA(-C)) sit ≈0.005–0.01 nats below the matched Pure-KDA control (D01) and ≈0.02 below the windowed hybrid (D02, 5.0B tokens). All arms train on one 8 GB GPU.

k-mer memory buys ≈0.005–0.01 nats over the matched pure-KDA control across the sweep, and the local window and curriculum are loss-neutral on top of it: the three arms are at near-*parity*. Whatever separates them elsewhere (§4.6, §4.2), it is not next-nucleotide modeling of megabase windows. This is the first of several places where perplexity carries no signal about the differences that matter downstream. Because each architecture was pretrained once, we treat pretraining-loss differences of this size, below the scale of expected run-to-run variance, as descriptive rather than statistically established.

4.2 Single-nucleotide resolution: Genomic Benchmarks and Nucleotide Transformer

Protocol sensitivity, measured on the same checkpoints. Because fine-tuning protocols in long-running studies drift — a better pooling head or a longer epoch budget is typically adopted for the newest arm first — we can quantify what such drift does to an architecture comparison, using this study's own history as the experiment. With Mnemos-DNA fine-tuned under an older protocol ([mean|last] head, 3 epochs) and Mnemos-DNA-C under the current one (bins+RC head, 6 epochs), the identical checkpoints produce 0.618 vs. 0.922 on splice sites — an apparent +30.4-point architecture effect. Under one matched protocol the same comparison is 0.740 vs. 0.733. On GUE, capping both arms' training sets identically (vs. full-data for one arm on 28 of 31 tasks) moves the apparent suite-mean gap from −5.3 points to −0.1. Every downstream number in this paper therefore uses one frozen protocol (6 epochs / 6,000 steps, 4,000/2,000 caps, bins+RC head, 2–3 fine-tuning seeds per cell), launched from a single queue script that records the full command line per row — including the random-init control, which is run under the same head as the arms it is compared against.

What the matched rerun shows (Figure 5, Table 3). Three things:

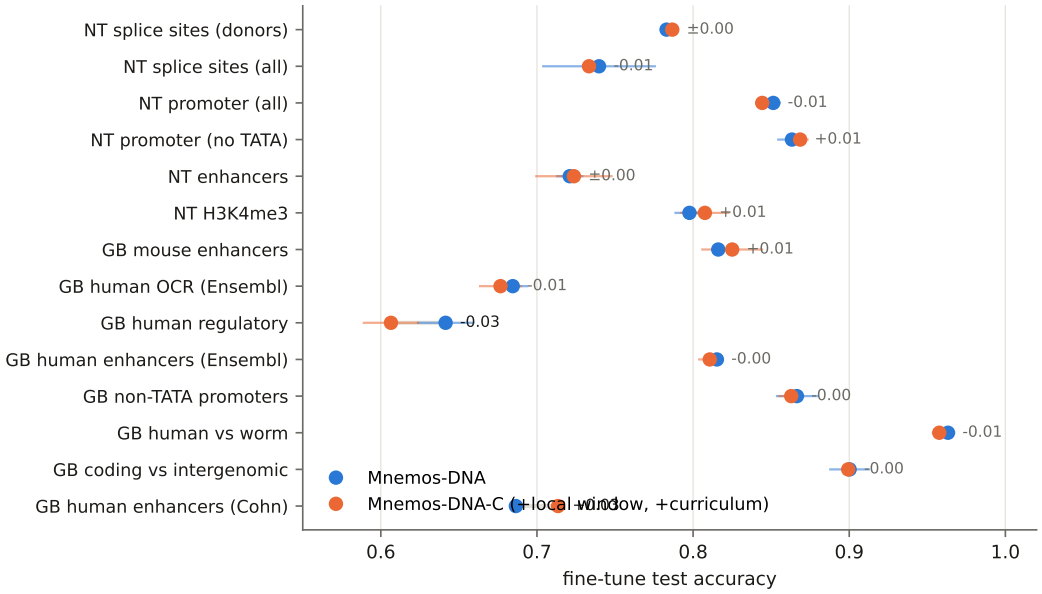


Figure 5: Full fine-tuning, Mnemos-DNA → Mnemos-DNA-C under the matched protocol, Genomic Benchmarks (8 tasks) and the 6-task Nucleotide Transformer subset; dots are seed means, whiskers ±1 sample std across fine-tuning seeds. No per-task difference exceeds 3.5 points and most sit inside the seed spread. Under an *unmatched* protocol the same checkpoints show spurious gaps of up to 30 points on this panel; see the protocol-sensitivity analysis below.

Task (full fine-tune, test accuracy)	Pure-KDA (D01)	Mnemos-DNA	Mnemos-DNA-C	random init
NT splice sites (donors)	0.779	0.783 ±0.001$^{(2)}$	**0.787 ±0.002**$^{(3)}$	0.897 ±0.005$^{(2)}$
NT splice sites (all)	0.713	**0.740 ±0.036**$^{(2)}$	0.733 ±0.003$^{(3)}$	0.832 ±0.002$^{(2)}$
NT promoter (all)	0.835	**0.851 ±0.004**$^{(2)}$	0.844 ±0.002$^{(3)}$	0.850 ±0.003$^{(2)}$
NT promoter (no TATA)	0.865	0.863 ±0.009$^{(2)}$	**0.869 ±0.005**$^{(3)}$	0.846 ±0.006$^{(2)}$
NT enhancers	**0.738**	0.721 ±0.008$^{(3)}$	0.724 ±0.024$^{(3)}$	0.735 ±0.010$^{(2)}$
NT H3K4me3	0.792	0.798 ±0.009$^{(3)}$	**0.808 ±0.015**$^{(3)}$	0.790 ±0.002$^{(2)}$
GB mouse enhancers	**0.831**	0.816 ±0.003$^{(2)}$	0.825 ±0.019$^{(3)}$	—
GB human OCR (Ensembl)	**0.694**	0.684 ±0.009$^{(2)}$	0.677 ±0.013$^{(3)}$	—
GB human regulatory	0.608	**0.641 ±0.018**$^{(2)}$	0.607 ±0.017$^{(3)}$	—
GB human enhancers (Ensembl)	0.799	**0.815 ±0.002**$^{(2)}$	0.811 ±0.007$^{(3)}$	—
GB non-TATA promoters	0.836	**0.867 ±0.013**$^{(2)}$	0.863 ±0.008$^{(3)}$	—
GB human vs worm	0.962	**0.963 ±0.004**$^{(2)}$	0.958 ±0.004$^{(3)}$	—
GB coding vs intergenomic	0.896	**0.900 ±0.012**$^{(2)}$	0.899 ±0.002$^{(3)}$	—
GB human enhancers (Cohn)	0.710	0.686 ±0.002$^{(2)}$	**0.714 ±0.002**$^{(3)}$	—

Table 3: Absolute per-task fine-tuned accuracy on the Nucleotide Transformer subset and all eight Genomic Benchmarks tasks under one matched protocol for every column (6 epochs / 6,000 steps, 4,000/2,000 train/test caps, [mean|last|8 bins]+RC head). Mnemos-DNA(-C) cells are mean ± sample std over fine-tuning seeds (per-cell n in superscript; Mnemos-DNA's seed-0 Genomic Benchmarks rows were archived by the harness's fingerprint guard, leaving n=2 there). Pure-KDA (D01) is a single seed. The random-init column fine-tunes the Mnemos-DNA architecture from scratch under the same head (2 seeds, Nucleotide Transformer only). All our arms are ∼2.3M-parameter models pretrained for 4.5B tokens on a single 8 GB consumer GPU; published HyenaDNA / Nucleotide Transformer / DNABERT-2 numbers on these suites come from larger models and far larger pretraining budgets, so any comparison to them is a comparison across compute budgets, not architectures.

- **The pretraining damage is real and survives the audit.** A randomly initialized Mnemos-DNA fine-tuned under the same head reaches 0.832 ± 0.002 on splice-sites-all and 0.897 ± 0.005 on splice donors, where the pretrained arms reach 0.71–0.79. On the position-sensitive splice tasks, the pretrained checkpoints (4.5B tokens at a fixed 131 K) are *worse* than no pretraining — by ∼9 and ∼11 points respectively. On the remaining Nucleotide Transformer tasks the pretrained arms are at or slightly above the random-init control, so the damage is specific to sub-kilobase positional grammar, not a general failure to transfer.
- **No repair under the matched protocol.** Mnemos-DNA-C's window-plus-curriculum moves splice-sites-all from 0.740 to 0.733 and splice donors from 0.783 to 0.787 — noise. Across all 14 tasks the arms differ by at most 3.5 points in either direction, and the suite means differ by ≤0.3 points. The +30.4 was the head and epoch budget, not the architecture.
- **The matched pure-KDA control lands in the same band.** D01, with no k-mer memory at all, is within ∼2 points of the memory arms on nearly every task. Whatever the fast-weight table contributes to recall (§4.6), these 200–600 bp classification tasks do not exercise it.

The damage is not specific to our architecture. To test whether the pretraining-vs-random-init inversion is a KDA quirk, we ran the same control on the released HyenaDNA-1M checkpoint: the pretrained artifact and a randomly initialized copy of the identical architecture (6.5M parameters, same tokenizer), both fine-tuned on the Nucleotide Transformer suite under the same matched protocol (random init: 2 seeds). The 1M-context pretrained checkpoint gains nothing over random init on any of the six sub-kilobase tasks, and on splice donors it falls clearly below (0.737 vs. 0.781 ± 0.009; splice-sites-all 0.655 vs. 0.689 ± 0.038, the same direction but within seed spread). The effect is milder than on our arms (4 points vs. 9–11) and rests on one pretrained seed, so we state it as: the signature generalizes — ultra-long fixed-context pretraining fails to help, and on the most position-sensitive task hurts, sub-kilobase grammar for an implicit-convolution architecture as well. Note these rows describe HyenaDNA's *1M-context* checkpoint under our matched protocol and caps; they are not comparable to HyenaDNA's published short-task numbers, which use its short-context (1 K) checkpoints — a deployment choice by its authors that is itself consistent with this finding.

Two ceilings are worth naming rather than burying. Nucleotide Transformer `promoter_all` is pinned at 0.84–0.85 for every arm *and* for the random-init control, while GC content alone yields 0.81: that is a d_{model}=128 capacity ceiling, not an architecture verdict. And the splice-site gap to random init is the sharpest open problem this study leaves: the curriculum-plus-window patch that was designed to close it does not.

4.3 Species classification across input lengths

Species identity is the task where more context monotonically helps: accuracy rises to 1.000 at 512 K–1 Mbp windows for Mnemos-DNA (Figure 6). The sweep embeds at batch 1, the deployment-relevant setting, where the externals reach 128–256 K; note that a batched embedding pipeline (8 sequences at once) instead drives them out of memory at 16–32 K, overstating the length gap by nearly an order of magnitude — the ceilings reported anywhere in this paper are single-sequence. Since all models saturate near 0.99 by 16 kbp, the accuracy axis is not where the arms separate — the length axis is, and that separation is a statement about deployable window size, discussed as such in §4.10.

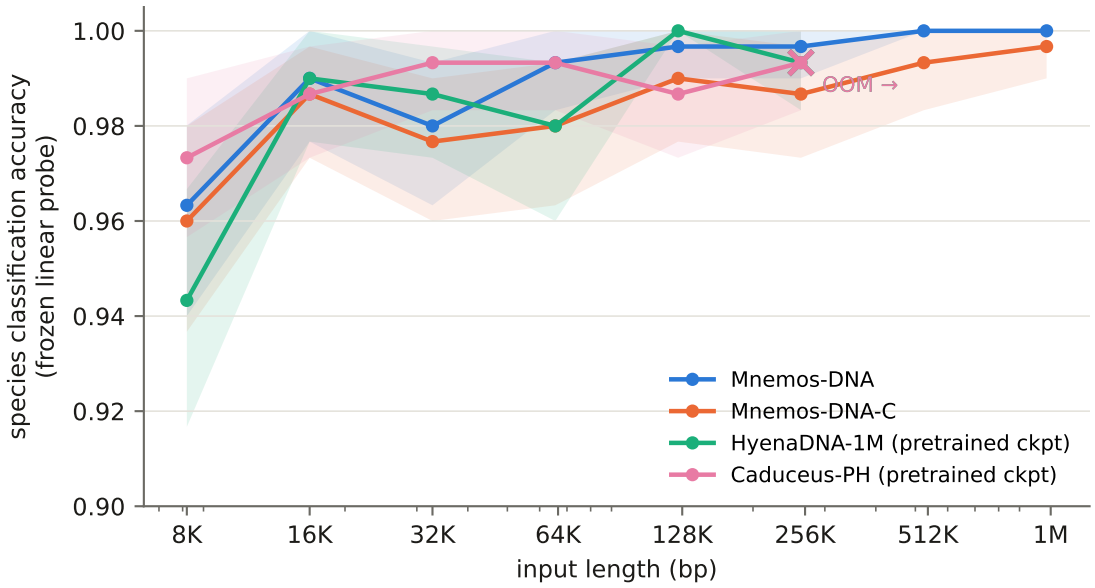


Figure 6: Species classification versus input length (frozen linear probe, batch-1 embedding, 5-way: human/mouse/zebrafish/fly/worm; shaded bands are 95% bootstrap CIs; × marks the last length before OOM). All models saturate near 0.99 by 16 K; the differentiator is how far each can go — Mnemos-DNA reaches 1 Mbp windows at 1.000 while the released external checkpoints stop at 256 K (Caduceus-PS at 128 K). The length ceilings compare released checkpoint artifacts, not architectures (§4.10).

What the embeddings look like. Figure 7 projects mean-pooled final-layer embeddings of 8 kbp windows with t-SNE [18], one panel per arm (D02, Mnemos-DNA, Mnemos-DNA-C). All three arms, trained only on *human* sequence with a causal next-nucleotide objective, organize five species into linearly separable clusters — which is why the frozen probe of Figure 6 works at all. The structure is consistent across arms: human, mouse and zebrafish (vertebrates) form adjacent clusters while fly and worm sit apart, i.e. the embedding geometry loosely recapitulates phylogenetic distance rather than being an arbitrary partition. The k-mer-memory arms (Mnemos-DNA(-C)) show modestly tighter vertebrate separation than the window-only D02, but all three are already probe-saturated at this length, so we present the panel as a qualitative account of *why* the probe saturates, not as an arm comparison.

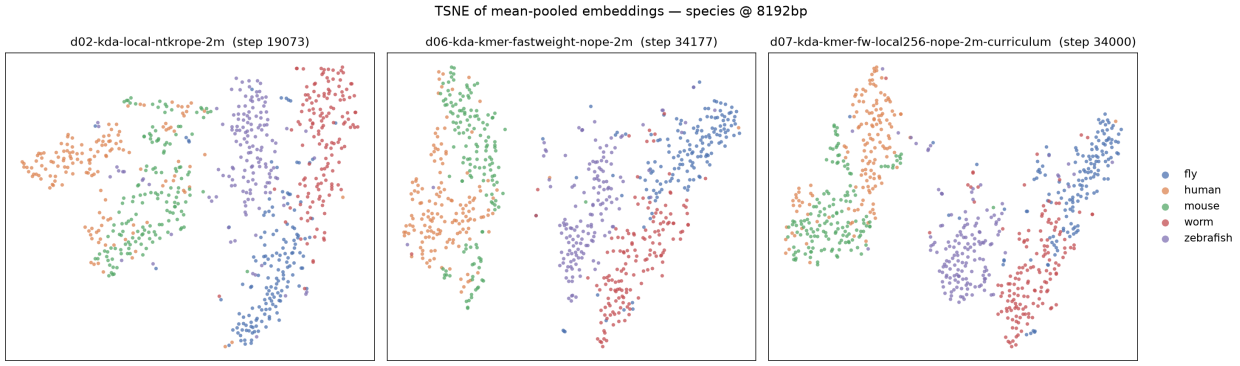


Figure 7: t-SNE of mean-pooled embeddings of 8,192 bp genome windows from five species (human/mouse/zebrafish/fly/worm), one panel per arm — D02 (window-only), Mnemos-DNA, Mnemos-DNA-C — all pretrained on human sequence only. Species separate without any species supervision, and vertebrates cluster together, mirroring phylogeny; this is the representation the frozen probe of Figure 6 reads out.

Task (probe, 8 K windows)	D01	Mnemos-DNA	Mnemos-DNA-C	D02	HyenaDNA-1M	Caduceus-PH	Caduceus-PS
GLRB VEP eQTL (acc)	0.664	**0.672**	0.625	0.664	0.586	0.586	**0.672**
GLRB enhancer (acc)	0.616	0.608	0.648	**0.664**	0.592	0.624	0.632
GLRB promoter (acc)	0.942	0.942	0.942	**0.950**	0.934	0.934	0.934
GLRB chromatin (mean acc)	—	0.931	0.937	0.929	0.933	0.937	**0.938**

Table 4: Genomics Long-Range Benchmark, frozen-backbone linear probes at 8 K context, same harness for every column; D01 is the matched pure-KDA control and D02 the window-only control of Table 2. A probe is a lower bound on all models equally. External checkpoints (right three columns) are ∼3× our parameter count and pretrained with orders of magnitude more compute than ours; typical 95% CIs span ±0.07 on eQTL/enhancer, so most differences here are individually indistinguishable — the honest summary is parity, reached at ∼2.3M parameters.

4.4 Long-range regulatory and variant-effect prediction

Table 4 gives the frozen-probe results on the Genomics Long-Range Benchmark [19] at 8 K windows: at ∼2.3M parameters our arms sit at parity with the external checkpoints on all four task groups, with most differences inside the bootstrap intervals. The more interesting result is what happens when the window grows. Where species identity rewards context, variant effect punishes it: every model degrades from 8 K to 32 K windows on GLRB VEP eQTL (Mnemos-DNA: 0.672 → 0.578; Table 4 reports the 8 K point). Diluting a SNP-centered signal with context is not free. "More context helps" is a per-task question, and this is the task where the answer is no — worth stating in a paper whose efficiency story is about long windows.

4.5 GUE at the matched protocol, and a regression that dissolved

Under the matched protocol the GUE ledger closes at zero: mean 0.750 (Mnemos-DNA) vs. 0.749 (Mnemos-DNA-C), with every per-task difference within ±3.8 points (Figure 8, Table 5). The unmatched-protocol picture had suggested two further conclusions; neither survives the matched rerun, and dissecting why is instructive.

`virus_covid` — under the unmatched protocol an alarming 0.657→0.177 regression that a learning-rate sweep could not recover — is explained: the 0.657 came from fine-tuning on the task's full 73,335-example training set, the 0.177 from the 4,000-example cap. Under matched caps both arms sit at 0.166–0.177 on this 9-class task, barely above the 0.124 majority baseline, with near-

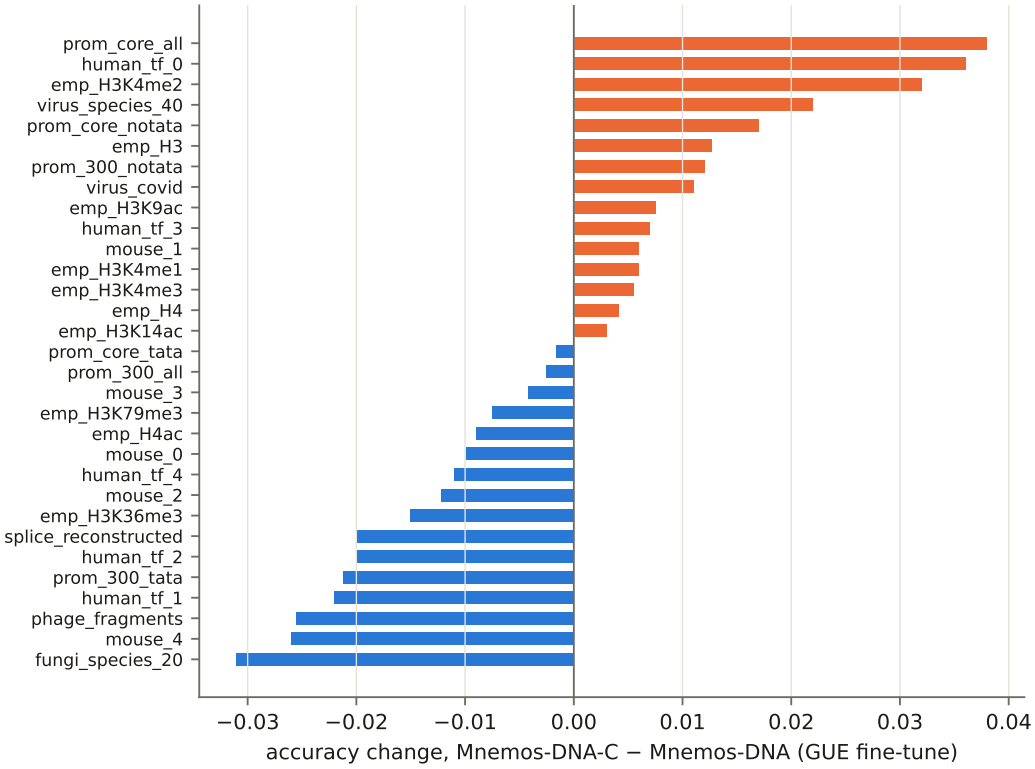


Figure 8: Per-task accuracy change on the 31-task GUE suite under the matched protocol (both arms: 4,000-example training cap, 6 epochs, bins+RC head, seed 0). The largest per-task difference in either direction is ±3.8 points and the suite means differ by 0.06 points — the arms are equivalent on GUE. Under an unmatched protocol (differently sized training sets on 28 of 31 tasks) the same checkpoints instead show a spurious −5.3-point mean loss for Mnemos-DNA-C, dominated by the yeast `emp_*` tasks.

Suite (fine-tune, mean acc)	#tasks	Pure-KDA (D01)	Mnemos-DNA	Mnemos-DNA-C	Δ(C−M)
Genomic Benchmarks [20]	8	0.792	**0.797**	0.794	−0.3
Nucleotide Transformer subset [7]	6	0.787	0.793	**0.794**	+0.1
GUE (DNABERT-2) [8]	31	—	**0.750**	0.749	−0.1

Table 5: Suite-level mean fine-tuned accuracy under the matched protocol (Genomic Benchmarks and Nucleotide Transformer means are over the seed means of Table 3; GUE is seed 0 for both arms; D01 was not run on GUE). GUE data is the `leannmlindsey/GUE` HuggingFace community mirror of the DNABERT-2 release (the official distribution cannot be fetched non-interactively), and should be cited as a community mirror. Note the GUE means are not comparable to published DNABERT-2 numbers: the 4,000-example training cap that makes the arms comparable to *each other* also trains on far less data than the official protocol on the larger tasks.

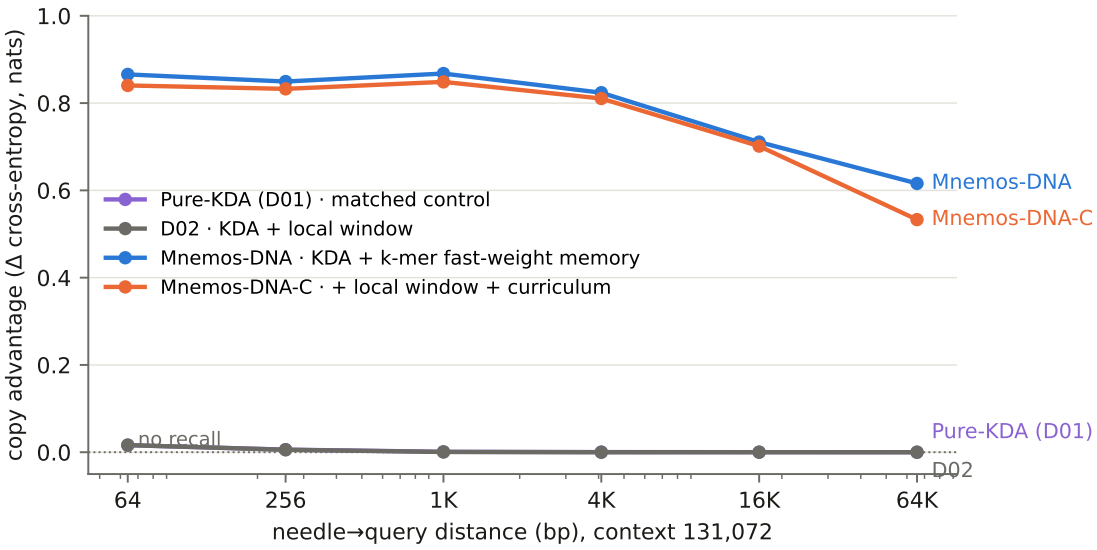


Figure 9: Long-range exact recall (context 131,072). A 256 bp segment of real hg38 sequence is planted and the model must lower its loss when the segment repeats a distance d later; the y-axis is the paired per-trial cross-entropy reduction versus a control (128 trials/point). The matched Pure-KDA control (D01) and the window-only hybrid (D02) are indistinguishable from zero beyond 64 bp. The k-mer fast-weight arms hold 0.53–0.87 nats of advantage out to 64 kbp, decaying smoothly with distance.

maximal prediction entropy. The correct statement is that *neither* arm learns SARS-CoV-2 variant classification from 4,000 examples; the learning-rate sweep was probing a data-budget artifact. The same applies to the `emp_*` yeast epigenetic-mark "degradation": it tracked training-set size, not architecture.

What survives is the dissociation: perplexity predicted none of this in either direction — the arms are 0.002 nats apart (§4.1), appear 5–30 points apart per task under an unmatched protocol, and are ≤4 points apart under the matched one. Downstream deltas of that size are within what *evaluation protocol variation alone* produces, which is itself this domain's restatement of "perplexity is not capability" — and a caution about how easily fine-tuned comparisons manufacture architecture stories.

4.6 Mechanism I: the recall bound

The mechanistic result resolves cleanly (Figure 9), and it now rests on a fully matched control: D01 shares Mnemos-DNA's positional scheme (NoPE), training context, token budget and data order, and differs only by the k-mer memory. Pure KDA's copy advantage is ≈0.02 nats at distance 64 and zero beyond — on genomic background it never performs measurable exact recall at *any* distance, so at this operating point the finite state is empirically insufficient to sustain exact recall over genomic background: not a distant cliff but a wall at distance zero. This is a statement about a $4 \times 32 \times 32$ state per layer, not about KDA at larger state dimensions. The window-only arm D02, rerun in copy mode for this figure, traces the pure-KDA curve point for point (0.016 at 64 bp, zero beyond): windowed local attention, despite exact token-level access inside its 512 bp span, does not by itself convert a repeated segment into a measurable copy advantage on hg38 background. The fast-weight memory (Mnemos-DNA) recovers 0.87 nats at short range and still holds 0.62 at 64 kbp. Mnemos-DNA-C sits slightly below Mnemos-DNA at long distances (0.53 at 64 kbp) — the

Arm	Params	State / layer (floats)	Copy advantage (nats) at distance 64	1 K	16 K	64 K	LM CE (nats/base)
D01 (Pure-KDA, 4×32^2)	2.25M	4,096	0.016	0.001	0.000	0.000	1.074
Pure-KDA 4× state (128^2)	2.25M	16,384	0.017	0.001	0.000	0.000	1.076
Pure-KDA 16× state (256^2)	8.96M	65,536	0.017	0.002	0.000	0.000	1.051
D06-poshash (position-hashed slots)	2.32M	16.8M	0.017	0.001	0.000	0.000	1.074
Mnemos-DNA (D06, 8-mer slots)	2.32M	16.8M	**0.866**	**0.868**	**0.711**	**0.616**	1.065

Table 6: Recall comes from content addressing, not from state or memory size. Repeat-copy probe of Figure 9 (context 131,072, 128 trials/point) and held-out LM cross-entropy at 131 K (4 M tokens), all re-measured in one run of the harness. 95% bootstrap intervals: every control's copy advantage is $\leq$0.022 at 64 bp and contains zero from 16 kbp on; Mnemos-DNA's lower bound is $\geq$0.56 at every distance. LM CE intervals are $\approx \pm 0.03$ (31 windows), so the 0.009-nat D06 advantage over D01 is within them. State per layer for the table arms counts the KDA state plus the fast-weight table. All arms trained on 4.5B tokens.

curriculum's short-sequence steps plausibly de-emphasize ultra-long carry — but the two memory arms are qualitatively identical. The controlled ablation indicates that the memory, rather than the tested local window, is responsible for the observed recall improvement.

Is it the memory's size, or its addressing? Three controls. The comparison above leaves two cheaper explanations open. *State size*: each D01 head's state is a single 32×32 matrix, and a larger pure-KDA state might recover recall without any table. *Memory size*: the k-mer table adds $4^8 \times 16 \times 16 \approx$16.8M floats of recurrent state per layer, and recall might come from that capacity regardless of how it is indexed. We trained three controls at the matched protocol (NoPE, 131 K context, same data order, cosine schedule; Table 6): pure KDA with 4× the per-layer state (one 128×128 head, parameter-matched to D01) and with 16× (one 256×256 head, d_{model}=256, the largest pure-KDA state whose training step fits the card at 131 K), and **D06-poshash**, Mnemos-DNA with its 4^8-slot table unchanged but each position's slot chosen by a hash of its *position* rather than of its last 8-mer — same parameters, same state, same write/read rule, no content addressing. (Randomly permuting the k-mer→slot map is not a control: it relabels slots and is equivalent to D06.)

Neither explanation survives. Quadrupling the state leaves the curve unchanged to three decimals, and sixteen-fold — with 4× the parameters, which buys a 0.023-nat LM improvement — moves the 1 kbp point from 0.001 to 0.002 nats. D06-poshash holds exactly Mnemos-DNA's 16.8M-float table and traces D01 point for point; its 1–3-substitution sweeps are likewise flat at zero, and its LM loss (1.0739) equals D01's (1.0738) to the fourth decimal, so a table the model cannot address by content is simply ignored. The recall in Figure 9 is therefore bought by *what the slot is keyed on*: two occurrences of the same 8-mer land in the same slot however far apart they are, which no position-indexed state, of any size we could train, provides. The claim stays bounded to what was measured — a 16× state at ∼9M parameters is still edge scale, and larger delta-rule models on language do show graded recall profiles rather than a wall [5, 3] — but within the range an 8 GB card can train, growing the pure-KDA state is not a substitute for the content-addressed memory.

The $0.82 \to 0.62$ nat decay between 4 kbp and 64 kbp is the beginning of the overwrite regime predicted in §3.2: at d=64 kbp = 4^8, a uniform sequence would have written the same slot about once between the two copies. The key-length sweep below makes this explicit.

k	Slots 4^k	Table state (all layers)	Copy advantage (nats) at distance 64	1 K	4 K	16 K	64 K	LM CE (nats/base)
— (D01, no table)	—	0	0.016	0.001	0.000	0.000	0.000	1.074
4	256	2 MiB	0.156	0.019	0.003	0.001	0.000	1.070
6	4,096	32 MiB	0.587	0.444	0.208	0.054	0.017	1.069
7	16,384	128 MiB	0.827	0.824	0.760	0.609	0.436	1.068
8 (Mnemos-DNA)	65,536	512 MiB	**0.866**	**0.868**	**0.824**	**0.711**	**0.616**	**1.065**

Table 7: Key-length sweep: recall reach scales with the address space. Same probe as Figure 9 (context 131,072, 128 trials/point) and held-out LM cross-entropy at 131 K (4 M tokens), all re-measured in one harness run. Table state is fp32 at inference, 8 layers $\times$ 4^k $\times$ 16$\times$16 $\times$ 4 B; parameters are 2.32M for every k. All k arms trained on 4.5B tokens (step 34,332; k=8 at step 34,177). 95% bootstrap intervals are $\leq \pm 0.06$ on every copy-advantage entry (Figure 10). The LM column spans 0.005 nats, inside its $\approx \pm 0.03$ interval: k is invisible to perplexity.

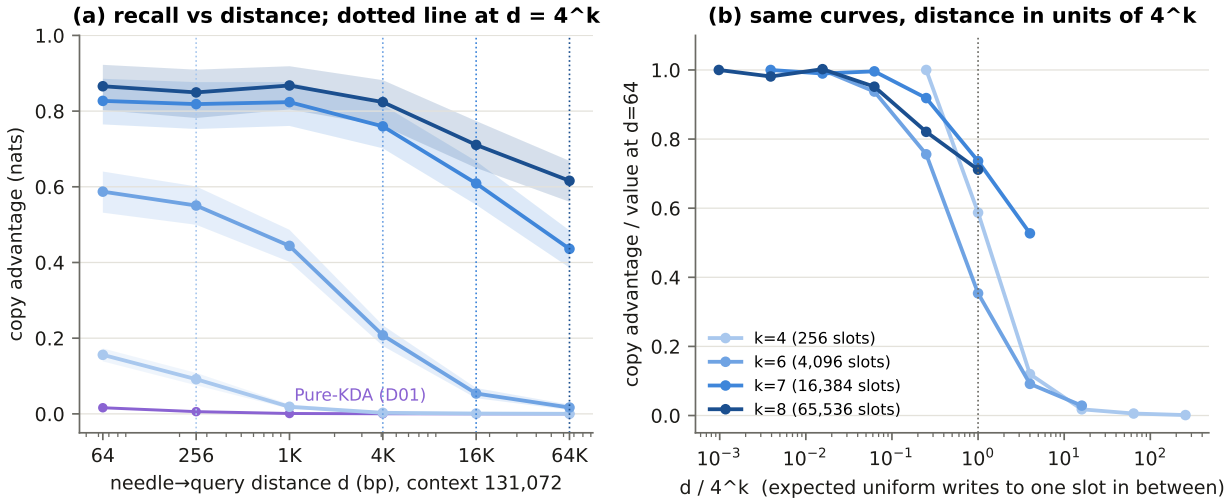


Figure 10: Recall fades once the repeat distance approaches 4^k. **(a)** Copy advantage against distance for $k = 4, 6, 7, 8$ (shaded: 95% bootstrap bands; dotted verticals at $d = 4^k$). **(b)** The same curves normalized by their d=64 value and plotted against $d/4^k$, the expected number of uniform-null writes to one slot between the two copies. The curves roughly line up. Each retains $\geq$90% of its short-range advantage while $d/4^k \leq 1/16$ and has lost 26–65% of it by $d/4^k = 1$. The alignment is approximate: k=6 fades earliest, and k=7 still keeps 53% at $d/4^k = 4$, where k=4 and k=6 keep $\leq$12%.

4.7 How much address space does recall need? A key-length sweep

Setup. If overwrite by intervening same-address writes limits recall, the reach of the memory should scale with the size of the address space, 4^k. We retrained Mnemos-DNA with $k \in \{4, 6, 7\}$ and compared it with the k=8 model. Everything else is held fixed (NoPE, 131 K context, 4.5B tokens, the same data order and schedule, 16 $\times$ 16 slots), so parameters are identical across k and only the table size changes (Table 7).

Results. The sweep supports the overwrite mechanism on three counts. *Reach scales with* 4^k. Recall holds at $\geq$90% of its short-range value while $d \lesssim 4^k/16$ and is clearly fading by $d = 4^k$ (Figure 10b). At k=4 the memory is spent within $\sim$1 kbp, at k=6 by 16 kbp, and only $k \geq 7$ carries a substantial advantage to 64 kbp. *Short-range recall also grows with* k: 0.16, 0.59, 0.83 and

0.87 nats at 64 bp. A 4-mer slot is written ∼500 times per 131 kbp window, so its content is an average over many unrelated contexts even when the copy is near. *Returns diminish.* Going from k=7 to k=8 multiplies the state by four and adds 0.04 nats at 64 bp and 0.18 at 64 kbp; going from k=6 to k=7 adds 0.24 and 0.42. Under the 1–3 substitution sweep every k loses a similar small per-substitution amount and keeps the shape of its curve (e.g. k=7: 0.83→0.70 at 64 bp, 0.44→0.36 at 64 kbp for m=0 → 3).

What this means for deployment. The 512 MiB state of Table 1 is not fixed. Choosing k sets it, and the right k follows from the repeat distance an application needs. For targets within ∼16 kbp, k=7 gives 0.61–0.83 nats in 128 MiB, and k=8 is only worth its cost when 64 kbp-scale recall matters. Perplexity cannot inform this choice, because the four memory rows of Table 7 sit within 0.006 nats of each other. The alignment in Figure 10b is only approximate. Real hg38 touches only 63% of 8-mer addresses (§4.8), so the effective address space, and thus the true scale, is smaller than 4^k by an amount that differs with k. We have not fitted that correction.

Degenerate motifs: the copy advantage survives point substitutions. Because the memory keys on *exact* 8-mers, the natural objection is that biological motifs are degenerate and a single SNP in a repeated segment could zero the mechanism out. We measured this directly: the probe re-run with m random point substitutions planted in the repeated 256 bp segment (`--needle-copy-mutations`; sequence identity 99.6% at m=1, 99.2% at m=2, 98.8% at m=3), for both memory arms at the full 131,072 context (Figure 11). For Mnemos-DNA the copy advantage falls from 0.87 to 0.74 nats at short range across m=0 → 3 and from 0.62 to 0.51 at 64 kbp; Mnemos-DNA-C behaves the same way (0.84→0.71 short range, 0.53→0.44 at 64 kbp). The cost is linear and small — ≈0.04 nats per substitution at short range, ≈0.03 at 64 kbp, with no threshold or collapse anywhere in the sweep — and the curve shapes are unchanged, so the decay with distance and the decay with divergence are independent. This is what the mechanism predicts: a substitution invalidates only the k=8 addresses overlapping it ($\leq$ 8 of 249 addressed positions per mutation), and at m=3 the remaining ≥90% of the segment's 8-mers still hit their slots. Exactness is required per *address*, not per *segment*. We claim this only for the tested low-divergence regime (up to 3 substitutions in 256 bp).

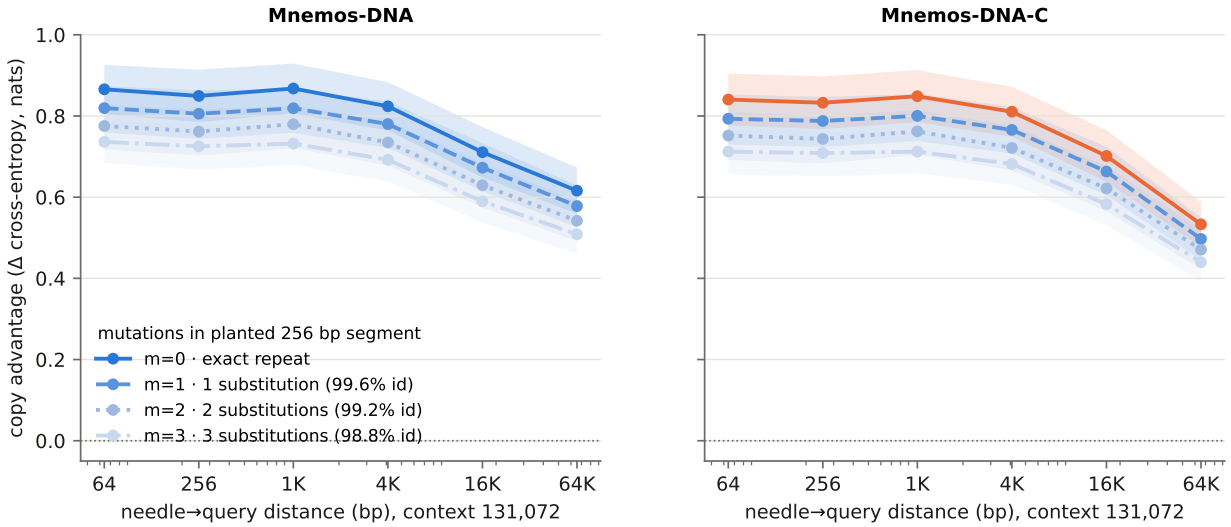


Figure 11: Recall under degenerate repeats (context 131,072, 128 trials/point, shaded 95% bootstrap bands). The repeat-copy probe of Figure 9 re-run with 0–3 random point substitutions planted in the repeated 256 bp segment, for Mnemos-DNA (left) and Mnemos-DNA-C (right). Each substitution costs ≈0.03–0.05 nats roughly uniformly across distances; the curves' shapes are unchanged, so the k-mer memory retains recall under the tested 1–3 substitutions between repeat copies — a substitution invalidates at most 8 of the segment's 249 8-mer addresses. All curves, m=0 through m=3, are measured on the main-suite checkpoints (step 34,177 for Mnemos-DNA, 33,900 for Mnemos-DNA-C).

Per 131,072 bp held-out window (Mnemos-DNA, 8 windows)	Measured	Uniform-sequence null
Addressed writes	131,056	131,056
Distinct slots written (of $4^8 = 65{,}536$)	41,342 (63.1%)	56,661 (86.5%)
Colliding writes	68.5%	56.8%
Writes per written slot (mean / p99 / max)	3.17 / 15 / 234	2.31 / 6 / —
Effective slots $\exp(H)$ of the write distribution	29,779 (45.4%)	56,661
Share of writes in the top-20 slots	0.98%	—
of which low-complexity (period ≤ 4)	0.50%	—
Mean write gate g_w, block 0 → block 6	0.226 → 0.865	—

Table 8: Slot utilization on real hg38. Addresses are a deterministic function of the sequence, so the first five rows are identical across all 8 layers; only the write gate differs by depth. Genomic non-uniformity costs capacity: 36.9% of the address space is never touched in a 131 kbp window, against 13.5% under a uniform null, and the write distribution's effective support is 45.4% of the space. The most-written slot is poly-T (234 writes; poly-A is second at 138), but the repeat tail is *narrow* — the top 20 slots take under 1% of all writes — so the capacity loss is broad non-uniformity, not a few repeat expansions. Deeper layers write much harder than shallow ones, which under Proposition 1 means a shorter retention horizon per slot near the output.

4.8 Mechanism II: how the 65,536 slots are actually used

Table 8 answers the capacity question empirically rather than from the pigeonhole bound. Three things stand out. *The space is under-used relative to its own null*: 63.1% of slots touched where uniform sequence would touch 86.5%. *Collisions are the norm*: 68.5% of writes land on an occupied slot, above the 56.8% a uniform null predicts and well above the 50% pigeonhole floor — which §3.2 argues is survivable, since a slot retains up to 16 near-orthogonal key directions. And *the*

popular intuition about repeats is only half right: poly-T and poly-A are indeed the top two slots by a wide margin, but the top 20 slots together carry under 1% of writes, so tandem repeats are not where the capacity goes. The gradient in write-gate strength with depth (0.23 at block 0 to 0.87 at block 6) is a learned division of labour — early layers accumulate over many occurrences of an 8-mer, late layers overwrite with the most recent one.

4.9 Mechanism III: which layer carries the splice-site signal

This probe was designed to ask whether Mnemos-DNA-C's local-attention block exposes splice-site structure — the mechanism the unmatched-protocol numbers had appeared to demand (§4.2). The measurement stands on its own as a map of where splice-site information lives in the stack, and it anticipated the correction: it found nothing special at the window. We fit an independent frozen linear probe on each of the 8 blocks' pooled representations (one embedding pass, identical splits) on Nucleotide Transformer `splice_sites_all`:

Block	0	1	2	3	**4 (local attn)**	5	6	7
Probe accuracy	**0.689**	0.687	0.666	0.654	**0.653**	0.668	0.673	0.672

There is no separability jump at block 4 — it is the worst block of the eight, and linear separability is highest at block 0 and declines through the middle of the stack. Read alongside the matched fine-tuning result, the pieces now agree: the window exposes no emergent linearly decodable splice-site structure, and under a matched protocol the fine-tuned numbers show no repair for it to explain. The declining-with-depth profile is itself informative — splice-site information is most linearly accessible at the *embedding-adjacent* blocks and is progressively traded away through the stack, consistent with 131 K-scale pretraining allocating depth to long-range statistics at the expense of local grammar, which is the same damage the random-init control exposes (§4.2).

4.10 Deployment feasibility of released checkpoints

This experiment evaluates the deployability of released checkpoint artifacts under an 8 GB constraint; it is not a parameter-matched algorithmic comparison. The external checkpoints are ∼3× our parameters (HyenaDNA-1M 6.5M, Caduceus-PH/PS 7.7M), were pretrained with far larger budgets, and are run as shipped without a chunked path, so every gap below confounds model size, implementation and architecture. Every other mention of the baselines in this paper refers back to this framing.

On the metric that matters for an edge deployment — the longest single sequence processable at all — our arms reach 1 Mbp at 5.6 GB, where the released checkpoints stop at 256 K (HyenaDNA-1M and -450K, OOM at 512 K) or 512 K (Caduceus-PH at 5.0 GB, Caduceus-PS at 7.6 GB; both OOM at 1 M) in the same harness on the same card; Figure 12 shows the batch-1 peak-VRAM curves, linear in length for every arm shown. Throughput is flat in length, as $O(N)$ requires (Figure 13).

Measurement protocol matters here as much as in fine-tuning: a batched embedding pipeline (8 sequences at once) yields apparent baseline ceilings of 16–32 K — an 8× overstatement of the gap. All ceilings reported in this paper come from single-sequence inference, the deployment-relevant measurement, under which our arms open windows 2–4× longer than the released baselines on this card.

A parameter-matched, chunking-matched comparison is a different experiment and we have not run it. One further qualification: the k-mer table makes our arms *less* memory-light than

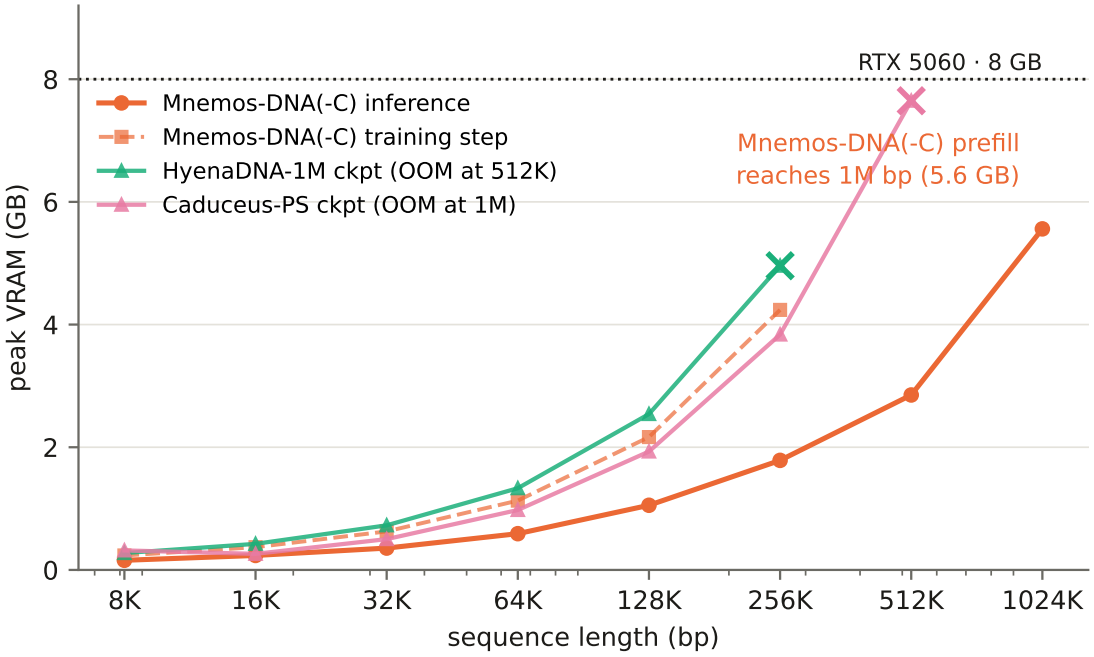


Figure 12: Peak VRAM versus sequence length (batch 1, bf16, single-sequence inference). Mnemos-DNA(-C) inference is linear and reaches 1 Mbp at 5.6 GB; a full training step fits through 256 K. The released baselines' own batch-1 curves are shown directly: HyenaDNA-1M reaches 256 K (5.0 GB) and goes out of memory at 512 K; Caduceus-PS reaches 512 K (7.6 GB) and goes out of memory at 1 M. Baselines run as shipped; our arms use a chunk-exact offload above 512 K (framing: §4.10).

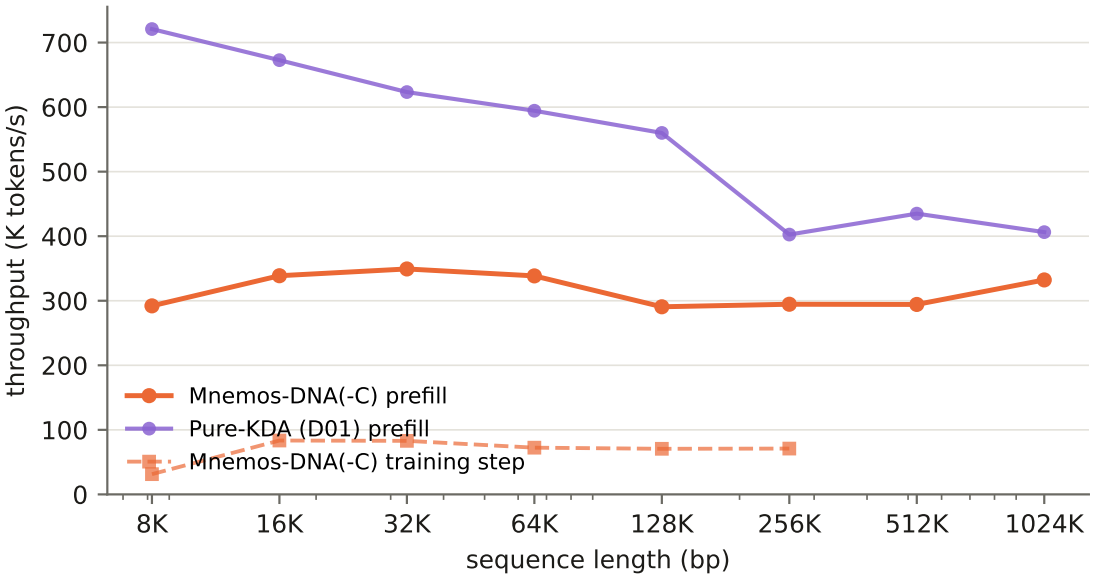


Figure 13: Throughput is flat in sequence length, as $O(N)$ requires: Mnemos-DNA(-C) prefill holds 250–350 K tok/s from 8 kbp to 1 Mbp, and a full training step holds ≈70 K tok/s. Against the matched Pure-KDA control (D01, 400–720 K tok/s) the k-mer table costs ≈2× prefill throughput on top of its VRAM cost (Table 1) — a real constant-factor price, though one that leaves the length-scaling flat (a prior comparison against the since-excluded D04 arm, whose inactive KV-cache path depressed its throughput, had understated this cost; Appendix B). Batch 1, bf16, one RTX 5060 (8 GB).

pure KDA (D01 runs a 1 Mbp forward in 5.5 GB natively and needs no offload anywhere in the sweep; Mnemos-DNA(-C) need the chunk-exact offload path above 512 K), which is the space cost of Table 1 showing up in practice.

5 Discussion and Limitations

Protocol drift can manufacture architecture "results," and we measured it doing so. Two apparent headline findings — a +30.4-point splice repair and a −5.3-point GUE cost — arose during this study from comparing arms fine-tuned under different heads, epoch budgets and training-set caps, protocol drift accumulated over months; both vanished under the frozen matched protocol. We consider quantifying this the most transferable finding in the paper: fine-tuned downstream comparisons between architectures are fragile to exactly the kind of incremental protocol improvement (a better pooling head, a longer budget) that gets applied to the newest arm first. Everything downstream in this version runs under one frozen protocol, launched from a single queue script that records the full command line per row.

The composite is not decomposed — but it no longer needs to be. Mnemos-DNA-C's two changes were introduced together, so any Mnemos-DNA-C-specific effect belongs to the pair. Under the matched protocol the pair's downstream effect is $\leq$1 point on every suite mean, its recall effect is a modest reduction at long range, and its LM effect is 0.002 nats; the curriculum-only control is now of secondary interest, relevant mainly to whether short-context validation parity (Figure 3) can be had for free.

Seeds and intervals. Fine-tuned numbers now carry 2–3 seeds per cell (Table 3); typical per-task seed spread is 0.2–2 points, which is why we treat sub-2-point differences as unresolved even where bootstrap intervals are narrow. Pretraining remains a single run per arm at one data order — pretraining-seed variance is unmeasured, and the bootstraps still cover split-sampling noise only. The large effects — the recall wall (0.6 nats), the random-init splice gap ($\sim$9 points), the OOM ceilings — are far outside both noise terms.

Exact recall is per address, not per segment. The 4^8 address space keys on exact 8-mers, and biology's motifs are degenerate — the objection we expected to be sharpest. The mutation sweep (§4.6, Figure 11) now bounds it for both memory arms: up to three point substitutions in the repeated segment cost $\approx$0.04 nats each, linearly and without a threshold, and do not change the shape of the recall curve. What remains open is heavier divergence — indels, rearrangements, and substitution rates high enough to break most of the segment's 8-mers rather than a few percent of them.

Capacity ceiling at d_{model}=128. Nucleotide Transformer `promoter_all` is pinned at 0.84–0.85 for every arm and for the random-init control (GC content alone yields 0.81). At the edge-scale operating point this study targets, some tasks are simply out of reach, and we report them as capacity ceilings rather than as architecture comparisons.

Baseline comparisons are about artifacts (§4.10). Where the released checkpoints match or beat us the reading is unremarkable; where we match them at 2.3M parameters the reading is parameter efficiency, nothing more.

What we would test next. An intervention that actually closes the splice-site gap to random init — the window-plus-curriculum patch did not, so the candidates are shorter or mixed pretraining lengths from step 0, or a pretraining objective term that protects local grammar; the remainder of the mutation sweep (indels and rearrangements); a window-size sweep of the local layer; a bf16 fast-weight table to halve the 512 MiB inference state; and, building on the key-length sweep, an address function that uses the 36.9% of address space hg38 never touches, e.g. mixed-k or frequency-balanced hashing, which should move each k's curve toward its uniform-null reach. One further control is still training at the time of writing: **D06-shared**, in which every write goes to a single 16×16 fast-weight matrix (memory without any addressing). It separates "content-addressed" from "any extra fast-weight memory". D06-poshash already argues for the former, and D06-shared will be reported when it finishes.

The k sweep is one seed per k. Its monotone ordering holds at every distance and the gaps between adjacent k are far larger than the bootstrap bands, but pretraining-seed variance per k is unmeasured, as for every arm here.

6 Conclusion

Gated delta-rule linear attention is a workable base operator for character-level genomic modeling at edge scale — but not alone. At this operating point's state capacity, pure KDA's finite-state recall bound is not a distant cliff on DNA; it is a wall at distance zero, and a windowed local-attention layer does not move it; neither does a 16× larger KDA state, nor a table of the same size indexed by position rather than content. A content-addressed k-mer fast-weight memory substantially improves recall on our probe — retaining it under the tested 1–3 substitutions in the repeated material — for ∼3% extra parameters, a ≈2× prefill-throughput constant, and a 4096× larger recurrent state. That state is tunable, since recall reach scales with the 4^k address space and fades once a repeat is about 4^k bp away, so k=7 keeps most of the recall in a quarter of the bytes. All of this preserves $O(N)$ scaling and megabase-length exact chunked inference on an 8 GB GPU; all three costs are accounted for rather than omitted. Its address space is used unevenly: two thirds of slots touched in a 131 kbp window, 68.5% of writes colliding, and a delta rule inside each slot that turns those collisions into a gated regression rather than a clobber. The fixed ultra-long pretraining context that makes such models impressive at length makes them *worse* at sub-kilobase grammar than their own random initialization — a failure our window-plus-curriculum patch did *not* repair, and which stands as the study's sharpest open problem. And fine-tuned downstream comparisons proved protocol-fragile: apparent 30-point architecture effects arise from head, budget and data-cap drift alone and vanish under one matched fine-tuning protocol, which we report as a result in its own right: at this scale, evaluation-protocol drift can produce larger architecture "effects" than architecture does. The trade-offs, not the wins, are the finding: recall, context length, and short-range grammar are three different budgets, and an architecture choice that spends one is spending it from somewhere.

A Pre-registered hypotheses and verdicts

The following hypotheses were stated in the study plan before the Mnemos-DNA(-C) comparison was run, so that outcomes could not be reframed after the fact. Each carries an explicit falsifier.

Hypothesis 1 (Deployment efficiency; high prior confidence)**.** *Under a fixed 8 GB VRAM ceiling, the $O(N)$ KDA arms will process strictly longer sequences than the released convolutional and SSM*

checkpoints, with peak memory growing linearly in sequence length. Falsifier: *any baseline matching or exceeding the KDA arms' maximum feasible context on the same card.* Test: *§4.10.*

Hypothesis 2 (Recall bound)**.** *Pure KDA's exact-recall accuracy will degrade past a distance threshold set by its finite state capacity; the k-mer fast-weight memory, which factors exact 8-mer recall out of that state, will hold up at longer range; a windowed local layer adds exact recall only inside its ∼256–511 bp span.* Falsifier: *pure KDA matching the memory-augmented arms at long range, or the memory arms degrading identically to pure KDA.* Test: *§4.6.*

Hypothesis 3 (Language-model loss; no prior stated)**.** *Pretraining perplexity across arms was pre-registered as genuinely uncertain, with no directional expectation.* Test: *§4.1.*

Hypothesis 4 (Downstream capability; dissociation expected)**.** *Downstream fine-tuned accuracy was pre-registered as uncertain, with the explicit caution that perplexity may not predict it. A Mnemos-DNA-C-specific sub-hypothesis, formed after observing Mnemos-DNA's splice-site failure and before Mnemos-DNA-C was trained: a local-attention window plus a length curriculum will repair sub-kilobase position-sensitive tasks that fixed-131 K pretraining damaged.* Falsifier: *Mnemos-DNA-C failing to close the gap to the random-initialization control on splice sites.* Test: *§4.2.*

B The excluded D04 arm, and a bf16 cautionary note

Earlier drafts used an arm labelled D04 as the pure-KDA control. An audit found that D04 was in fact a KDA variant with an auxiliary evicted-KV exact cache whose 32 scalar cache gates, stored in bf16, sat at *exactly* their -4.0 initialization (sigmoid $= 0.018$) after the full 5.0B-token run: bf16's rounding granularity around small gradient updates froze them, so the cache path was inactive for the entire run while still costing throughput (its 80–98 K tok/s was the source of a prior, incorrect "the k-mer table is throughput-free" comparison). D04 was excluded and replaced by D01, a true pure-KDA arm trained under the fully matched protocol (NoPE, 131 K context, 4.5B tokens, same data order). The general lesson is the one already noted in §3.1: scalar gate parameters must be kept in fp32; every arm in this paper does so, and any arm trained before that fix should have its gate parameters checked against their initialization before its numbers are trusted.

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Hypothesis	Pre-registered prediction	Verdict	Evidence
H1 deployment efficiency	KDA arms exceed released baselines' max context on 8 GB	**Supported**, with a shrunk margin	1 Mbp vs. 256 K–512 K before OOM at batch 1 (Figs. 12, 6); an earlier 32 K/16 K figure was a probe-batch confound; confounds model size (∼3×)
H2 recall bound	Pure KDA degrades past a distance threshold; memory arms hold longer; a local window adds recall only inside its span	Supported, stronger than predicted	Matched pure KDA (D01) has *no* recall at any distance, nor do 4×/16×-state pure KDA or a position-hashed table of Mnemos-DNA's size (Table 6); the window-only arm (D02) is identical to D01; memory arms hold to 64 kbp (Fig. 9) and under the tested 1–3 substitutions (Fig. 11); the memory's own reach scales with 4^k (Table 7)
H3 LM loss	No direction stated	Parity	All arms within 0.01 nats; flat 8 K–1 M (Figs. 3, 4)
H4 downstream	Perplexity–capability dissociation; window+curriculum repairs sub-kb tasks	Dissociation supported; **repair sub-hypothesis falsified**	Under the matched protocol Mnemos-DNA-C does not close the gap to the random-init control on splice sites (0.733 vs. 0.832), so the stated falsifier fired. An earlier apparent +30.4-point repair was a fine-tuning-protocol artifact (§4.2); the GUE regression reported alongside it dissolved under the same audit

Table 9: Verdicts on the pre-registered hypotheses. "Supported" means the falsifier stated above did not occur — not that the proposed mechanism is confirmed. H4's repair sub-hypothesis is the study's one falsified prediction, and it was briefly "supported" by numbers that a protocol audit later invalidated — the sequence of events §5 argues is itself worth reporting.

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